

Comparative Analysis of Image Cleaning Processing Techniques for Imaging Atmospheric Cherenkov Telescope Data

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Abstract—Imaging Atmospheric Cherenkov Telescopes (IACTs) detect very-high-energy γ -rays through the Cherenkov light emitted by extensive air showers in the Earth’s atmosphere. Since the recorded camera images are contaminated by Night Sky Background (NSB) photons and detector noise, image cleaning is one of the most critical stages of the data reconstruction pipeline. An inefficient cleaning algorithm may either remove genuine shower pixels or retain excessive background noise, ultimately degrading the reconstruction accuracy and the telescope sensitivity.

Inspired by recent developments in time-based image cleaning for IACTs, this work presents a comparative study of several image-processing techniques for NSB suppression. Synthetic Cherenkov camera images were generated following the classical Hillas parameterization, allowing the exact shower morphology to be known for every event. Thresholding, morphological operators and clustering techniques are evaluated in terms of their ability to preserve the shower while rejecting background pixels.

The obtained results demonstrate that classical image-processing methods provide an effective framework for Cherenkov image cleaning and constitute a suitable basis for more advanced time-dependent approaches.

Index Terms—Gamma-ray astronomy, Imaging Atmospheric Cherenkov Telescopes, image processing, image cleaning, Night Sky Background.

I. INTRODUCTION

VERY-high-energy (VHE) gamma-ray astronomy studies some of the most energetic phenomena in the Universe, including supernova remnants (SNR), pulsar wind nebulae (PWNe), active galactic nuclei (AGN) and gamma-ray bursts (GRB). Since gamma rays cannot be directly detected from the ground, their presence is inferred from the extensive air showers they generate after interacting with nuclei in the Earth’s atmosphere.

The charged secondary particles produced during the shower – created due to the interaction between the first high-energy particle and the atmosphere – propagate faster than the speed of light in air and emit Cherenkov radiation, generating a short flash of bluish light lasting only a few nanoseconds. This light is recorded by Imaging Atmospheric Cherenkov Telescopes (IACTs), which reconstruct the direction and energy of the primary gamma ray from the recorded camera image [1].

Current efforts in VHE astronomy are centred on the Cherenkov Telescope Array Observatory (CTAO), the next-generation ground-based gamma-ray observatory. CTAO is expected to provide unprecedented sensitivity and angular resolution, enabling the detection of fainter astrophysical sources

and improving our understanding of particle acceleration mechanisms and the origin of cosmic rays [2].

One of the main challenges of IACT observations is that the Cherenkov signal is recorded together with a significant amount of Night Sky Background (NSB) photons, electronic noise and accidental triggers. Consequently, the recorded images contain many pixels that do not belong to the physical shower. Removing this background while preserving the true shower morphology constitutes one of the most critical stages of the reconstruction pipeline. Since all subsequent reconstruction algorithms rely on the cleaned image, the quality of the image-cleaning procedure directly impacts the overall telescope performance.

Recent work by Čelić *et al.* [3] proposed a novel optimization strategy for image cleaning based on the final telescope sensitivity rather than intermediate image-quality metrics. Inspired by this work, the present project focuses on the image-processing stage of the reconstruction chain.

II. DATA

Image-cleaning algorithms operate on calibrated camera images (DL1), where each pixel contains the recorded Cherenkov intensity before any cleaning procedure has been applied. Although several public datasets from Cherenkov telescopes are available, most correspond to DL3 reconstructed events, for which the image-cleaning stage has already been completed. Consequently, they are not suitable for evaluating image-cleaning algorithms.

The reference work by Čelić *et al.* [3] employs detector-specific Monte Carlo simulations using CORSIKA [4] together with the H.E.S.S. analysis chain. Reproducing the complete simulation pipeline is beyond the scope of the present work and, more importantly, is not required to investigate the image-processing problem itself.

Instead, synthetic Cherenkov camera images are generated following the parameterization introduced by Hillas [5]. Each event is represented by a two-dimensional elliptical Gaussian distribution whose position and orientation are randomly varied, while eccentricity is kept at a constant value, and intensity is scaled linearly in order to compare the different methods. Random NSB photons and detector noise are then added, following a Gaussian distribution with mean 0 and standard deviation equal to 1.65 p.e. as [3] states to reproduce realistic observation conditions.

Furthermore, for the usage of the DBSCAN method – which will be later introduced – the algorithm needs the time-dependence of the pixels of the array. This is that, the image

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recorded by the IACT is composed of two arrays: the one with the maximum intensity recorded that we have just explained; and the one with the time of arrival of the signal. This event has a standard deviation of ~ 1 ns, so if we state that the time of arrival of the signal is at $t' = 15$ ns, then the time dependence of the gaussian bulb zone will be represented by a gaussian distribution with mean t' and a standard deviation 1 ns. On the other hand, noise arrives uniformly during all of the recording, so if we suppose that the recording lasts 40 ns then the temporal distribution of this noise is uniformly represented in this time interval.

Synthetic datasets are widely employed for the development and validation of image-processing algorithms because they provide complete control over the image properties while preserving the relevant physical characteristics.

Unlike previous work, which evaluates image-cleaning algorithms within detector-specific reconstruction pipelines, the present work isolates the image-processing stage and studies it independently. By generating synthetic Cherenkov images with known ground truth, we obtain a controlled benchmark that allows objective comparisons between different algorithms while remaining independent of any particular telescope implementation.

III. METHODOLOGY

This section presents the performance of the different image-processing techniques investigated throughout this work. Each method is evaluated using the same synthetic dataset in order to compare its capability to suppress NSB while preserving the morphology of the Cherenkov shower.

Four different image-cleaning strategies were implemented and compared:

- **Tail-cut algorithm:** after deleting the pixels below $3\sigma_{noise}$, the method relies on two thresholds. Pixels with a signal above the lower threshold are retained only if they are adjacent to a pixel exceeding the higher threshold. This condition is bidirectional: pixels above the higher threshold are retained if they are adjacent to a pixel with a signal above the lower threshold. For this work, we will be using the 0916NN2 configuration, where the higher threshold is 16 p.e. and the lower one is 9 p.e.
- **Morphological approach:** based on opening and closing operations, introduced during course and provided by the Skimage library.
- **K-Means clustering** [6]: introduced during the course, we will be using it to separate the image in two clusters: noise and signal. Typically kmeans, would be used just with the intensity, but the result is a global thresholding not much interesting. In this case, we have included the information of the position, scaled with the intensity and moduled by a paramter that gives the right weight to the position in front of the intensity.
- **DBSCAN clustering** [7]: Density Based Spatial Clustering of Applications with Noise (DBSCAN) is an efficient clustering method that operates without predefined cluster counts. It relies on the inclusion of the temporal information of the shower. There are two possible paradigms:

TIME4D and TIME3D, depending on the inclusion or not of a dimension that represents the logarithmic variation of intensity. We will be using the second one.

Each method was applied to simulated Cherenkov images with increasing shower intensity, ranging from 5 to 50 photoelectrons (p.e.), while keeping the NSB noise fixed. Two quantitative metrics were used throughout the comparison:

- **Signal retention:** fraction of the original shower intensity preserved after the cleaning procedure.
- **Background survival:** fraction of background pixels that remain after cleaning.

In addition, representative reconstructed images were visually inspected in order to evaluate the preservation of the shower morphology.

IV. RESULTS

A. Visual comparison

Figure 1 presents representative examples for low (10 p.e.), intermediate (30 p.e.) and high (50 p.e.) intensity showers.

At low intensities, the tail-cut algorithm is unable to recover the shower because most pixels fall below the predefined thresholds. In contrast, the morphological approach preserves the core of the shower while efficiently suppressing isolated noise pixels. K-Means is capable of identifying the shower region but leaves a noticeable amount of residual background due to the overlap between signal and noise in the feature space. Finally, DBSCAN successfully identifies the shower by exploiting the spatial and temporal coherence of neighbouring pixels, producing the most complete reconstruction among the four methods.

As the shower intensity increases, all methods improve substantially since the signal becomes easier to distinguish from the background. Nevertheless, important differences remain.

Tail-cut continues to remove part of the shower edges because of its fixed thresholds, whereas the morphology-based method preserves a larger fraction of the original image.

K-Means provides compact shower masks with very little background contamination.

The DBSCAN reconstruction remains the closest to the original shower, retaining both the central core and a significant fraction of the lower-intensity pixels surrounding it.

B. Signal retention

Figure 2 summarizes the signal retention obtained by each algorithm as a function of the shower intensity.

The tail-cut algorithm exhibits the lowest performance for faint showers, retaining almost no signal below approximately 15 p.e. This behaviour is expected since the algorithm relies on fixed intensity thresholds that become too restrictive when the shower is weak. Above 20 p.e., the retained signal increases steadily, reaching approximately 78% at the highest tested intensity.

Morphological cleaning performs significantly better at low and intermediate intensities. Even for showers around 10 p.e., approximately 40% of the signal is preserved, while for intensities above 30 p.e. the retention stabilizes around 85–90%.

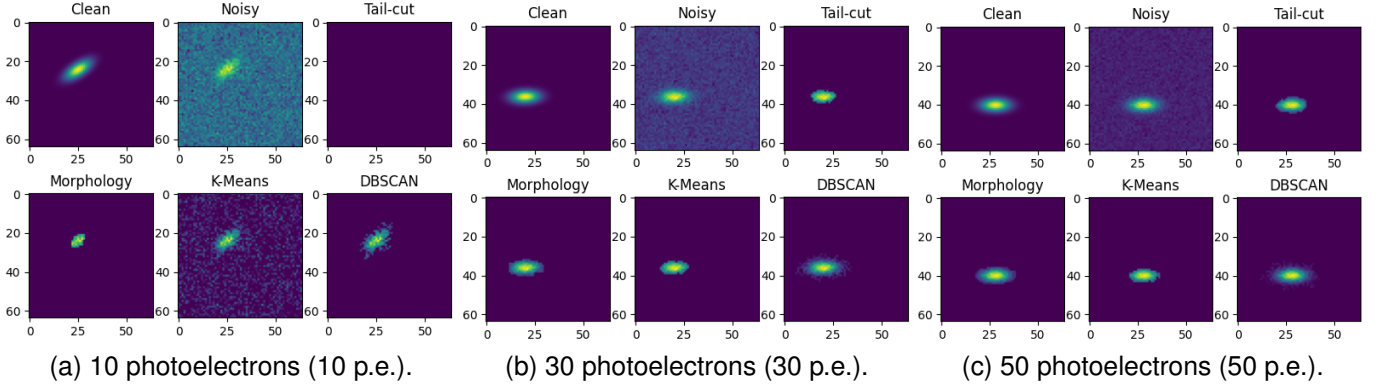


Fig. 1. Comparison of the denoising and background-removal methods at three signal intensities. Each subfigure shows the clean image, the noisy input, and the outputs obtained using tail-cut, morphology, K-means, and DBSCAN filtering. (a) 10 p.e., (b) 30 p.e., and (c) 50 p.e.

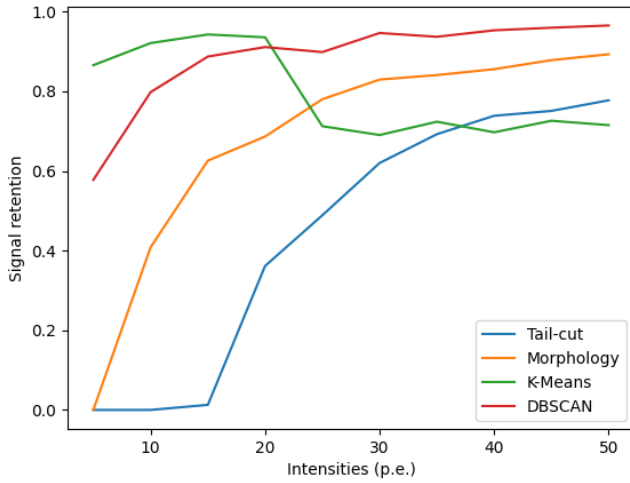


Fig. 2. Signal retention as a function of signal intensity for the tail-cut, morphology, K-means, and DBSCAN methods. Higher values indicate better preservation of the signal after background removal.

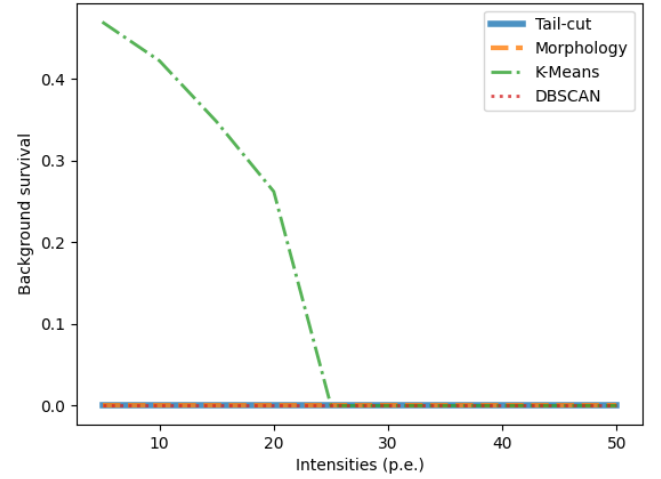


Fig. 3. Background survival as a function of signal intensity for the tail-cut, morphology, K-means, and DBSCAN methods. Lower values indicate more effective suppression of background noise.

K-Means clustering shows a different behaviour. It achieves excellent signal retention for weak showers, exceeding 90% between 10 and 20 p.e., but its performance gradually decreases for higher intensities, stabilizing around 70%. This reduction is mainly caused by the clustering algorithm assigning part of the shower periphery to the background cluster as the shower becomes more extended.

Among all methods, DBSCAN provides the highest signal retention over the entire intensity range. Even for the faintest events, more than half of the original signal is preserved, while intensities above 20 p.e. consistently retain more than 90% of the shower. This behaviour demonstrates the advantage of incorporating neighbourhood information instead of relying exclusively on pixel intensity.

C. Background rejection

The background survival for all methods is shown in Figure 3.

Tail-cut, morphological filtering and DBSCAN remove almost all background pixels for every tested intensity, resulting in negligible background survival.

K-Means behaves differently at low intensities, where nearly half of the background pixels are incorrectly classified as belonging to the shower. As the shower intensity increases, however, the separation between signal and background becomes progressively clearer, and the background survival rapidly decreases, reaching values close to zero above approximately 25 p.e.

These results indicate that K-Means is considerably more sensitive to the overlap between the statistical distributions of signal and noise than the remaining methods.

V. DISCUSSION

Although the simulated dataset does not reproduce every detector effect present in real IACT observations, it captures the characteristics that are most relevant to the image-cleaning problem: elongated shower morphology, random orientation, variable intensity and background contamination.

The main advantage of using synthetic data – which improves our analysis with respect to the literature followed – is the availability of the exact ground-truth shower for every event. This enables an objective comparison between different

TABLE I
QUALITATIVE COMPARISON OF THE IMPLEMENTED IMAGE-CLEANING METHODS.

Method	Signal retention	Background rejection	Main limitation
Tail-cut	Low for weak showers	Excellent	Fixed intensity thresholds
Morphology	High	Excellent	Depends on the structuring element
K-Means	Highest for weak showers	Poor at low intensities	Sensitive to cluster overlap
DBSCAN	Highest overall	Excellent	Requires tuning of ε and MinPts

cleaning techniques, something that is generally not possible with observational data.

While the reference work investigates detector-specific time-based cleaning algorithms, the present study explores several generic image-processing techniques that require no detector-specific calibration and could therefore serve as alternative preprocessing strategies.

A. Comparison between methods

Tail-cut relies exclusively on intensity thresholds and therefore assumes that signal pixels are brighter than background pixels. This assumption progressively breaks down for low-energy showers, where Cherenkov pixels exhibit amplitudes comparable to NSB fluctuations. Consequently, the algorithm sacrifices signal preservation in order to maintain low false-positive rates.

Morphological operators introduce an implicit spatial prior by assuming that genuine showers form connected structures whereas NSB noise appears as isolated pixels. This assumption is valid for the simulated data and explains the substantial improvement over pure thresholding. However, excessively aggressive structuring elements could distort the shower geometry, particularly for fragmented or partially contained events.

K-Means minimizes within-cluster variance under the assumption that clusters are approximately spherical in feature space. Since signal intensity gradually decreases towards the shower edges, boundary pixels overlap statistically with the NSB distribution. Consequently, these pixels are frequently assigned to the background cluster, explaining the reduction in signal retention observed for high-intensity showers.

DBSCAN does not require specifying the number of clusters and naturally rejects isolated points as noise. This behaviour closely matches the physical characteristics of Cherenkov showers, which are spatially connected structures embedded within randomly distributed NSB pixels. Consequently, DBSCAN represents a physically motivated approach rather than simply a statistical classifier.

VI. CONCLUSION

Image cleaning is a fundamental stage of the analysis pipeline of IACTs, since the quality of the reconstructed gamma-ray event strongly depends on the capability to suppress the NSB while preserving the genuine Cherenkov signal.

Inspired by the image-cleaning strategy proposed by Čelić *et al.* [3], this work investigated several classical image-processing techniques together with clustering methods using synthetic Cherenkov camera images. The adopted simulation reproduces the essential characteristics of gamma-ray showers while providing an exact ground truth that enables an objective

evaluation of each algorithm, which remains as an improvement with respect to the literature consulted.

The obtained results demonstrate that classical image-processing techniques constitute an effective framework for Cherenkov image cleaning. Morphological operations and clustering methods successfully reduce background contamination while preserving the overall shower morphology, illustrating how image-processing tools can be naturally applied to modern gamma-ray astronomy.

Although our methodology cannot replace detector-level validation using complete Monte Carlo simulations, it provides exact ground truth for every shower, enabling quantitative measurements of signal preservation and background rejection that are impossible with real observational data.

The two approaches should therefore be regarded as complementary rather than competing. Our framework is suitable for rapid algorithm development and benchmarking, whereas the methodology of Čelić *et al.* remains necessary for final validation on realistic telescope data.

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